SOC Analyst Assignment: Week 3 Report

A.Theoretical Foundation

1. Advanced Log Analysis Concepts

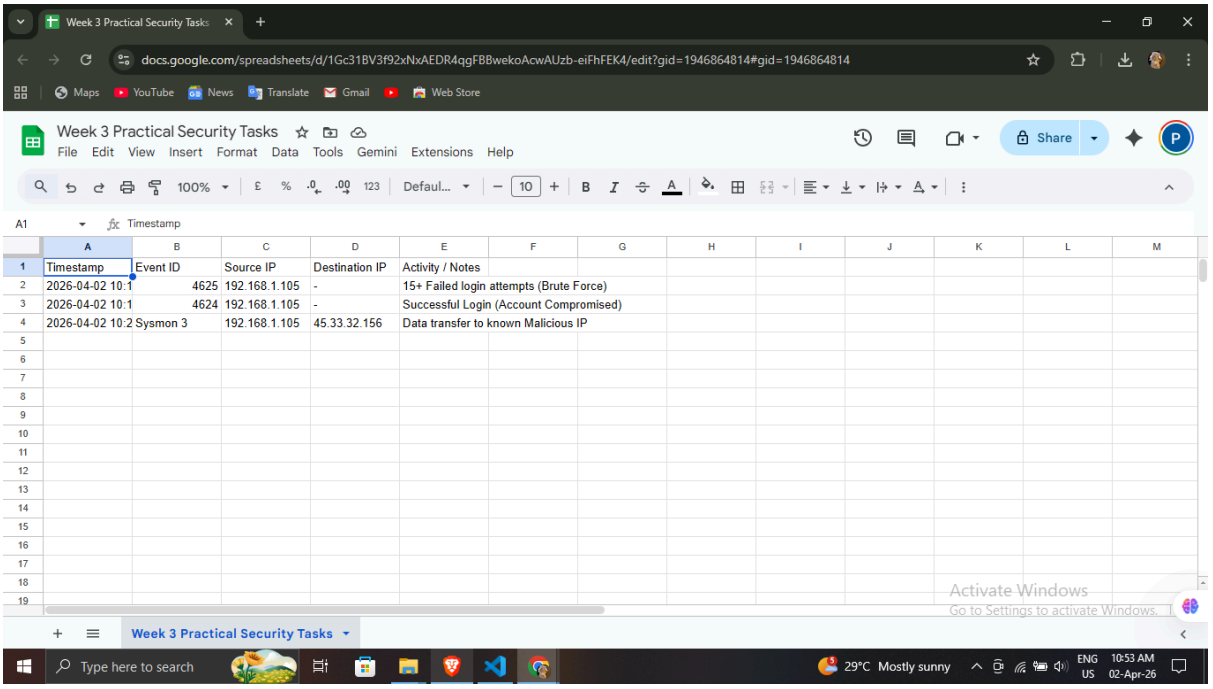
- **Log Correlation:** The process of aggregating logs from diverse sources (Firewalls, Windows Event Logs, Sysmon) to identify complex attack patterns.
- **Anomaly Detection:** Identifying deviations from established baselines, such as a user logging in at 3:00 AM or a sudden spike in outbound data transfers.

2. Threat Intelligence & Escalation

- **STIX/TAXII Standards:** STIX is the standardized language used to describe threat information, while TAXII is the protocol used to exchange that information.
- **Incident Escalation (NIST SP 800-61):** Includes Triage (Tier 1), Investigation (Tier 2), and Advanced Forensics/Briefing (Tier 3).

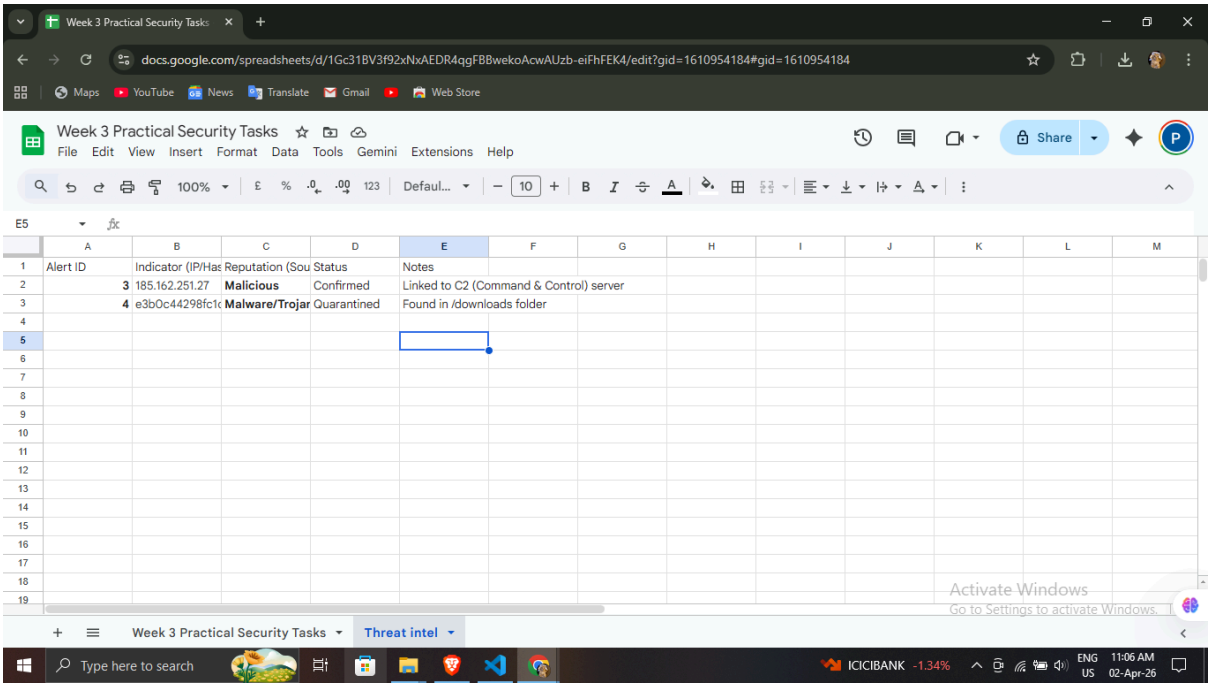
B.Practical Application (Logs & Intel)

1. Advanced Log Analysis (Correlation Table)



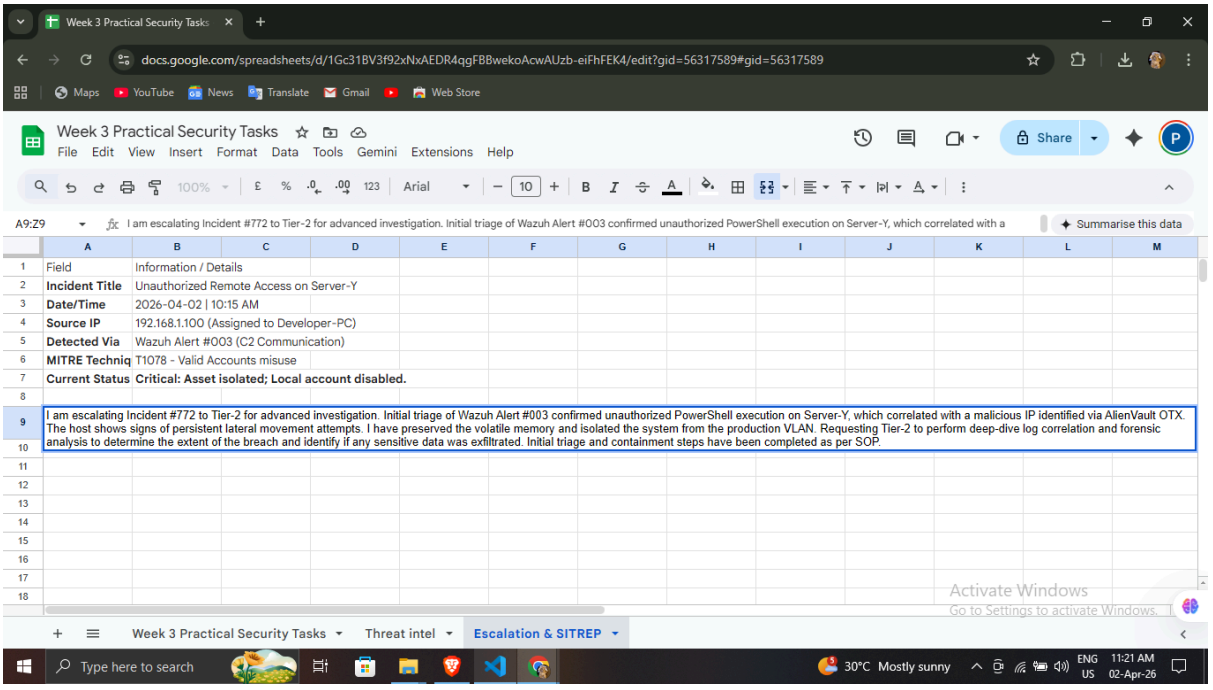
Timestamp	Event ID	Source IP	Destination IP	Activity / Notes
2026-04-02 10:1	4625	192.168.1.105	-	15+ Failed login attempts (Brute Force)
2026-04-02 10:1	4624	192.168.1.105	-	Successful Login (Account Compromised)
2026-04-02 10:2	Sysmon 3	192.168.1.105	45.33.32.156	Data transfer to known Malicious IP

2. Threat Intelligence Integration



Alert ID	Indicator (IP/Has	Reputation (Sou	Status	Notes
3	185.162.251.27	**Malicious**	Confirmed	Linked to C2 (Command & Control) server
4	e3b0c44298fc1c	**Malware/Trojar**	Quarantined	Found in /downloads folder

3. Incident Escalation Practice (SITREP)



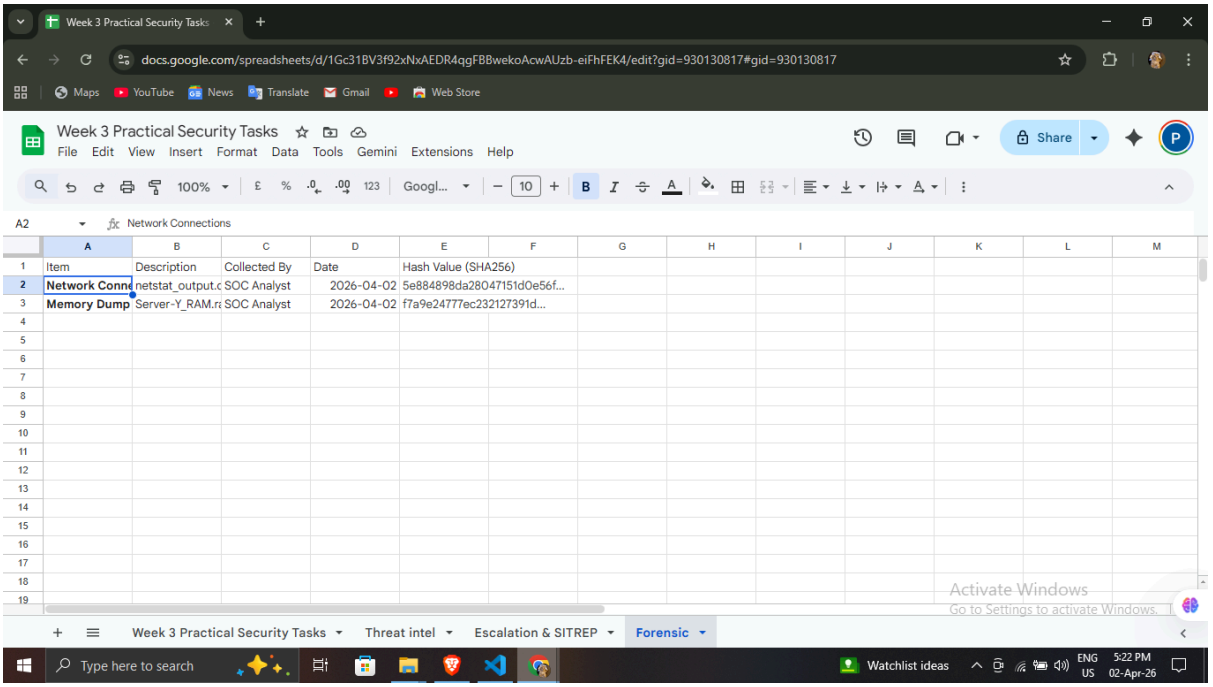
Field	Information / Details
Incident Title	Unauthorized Remote Access on Server-Y
Date/Time	2026-04-02 | 10:15 AM
Source IP	192.168.1.100 (Assigned to Developer-PC)
Detected Via	Wazuh Alert #003 (C2 Communication)
MITRE Techniq	T1078 - Valid Accounts misuse
Current Status	**Critical: Asset isolated; Local account disabled.**

I am escalating Incident #772 to Tier-2 for advanced investigation. Initial triage of Wazuh Alert #003 confirmed unauthorized PowerShell execution on Server-Y, which correlated with a malicious IP identified via AlienVault OTX. The host shows signs of persistent lateral movement attempts. I have preserved the volatile memory and isolated the system from the production VLAN. Requesting Tier-2 to perform deep-dive log correlation and forensic analysis to determine the extent of the breach and identify if any sensitive data was exfiltrated. Initial triage and containment steps have been completed as per SOP.

Escalation Summary: High-priority alert (ID: 003) involves multiple failed login attempts followed by a successful bypass on a critical asset. Post-exploitation logs indicate a PowerShell execution communicating with a blacklisted external IP. I have initialized a 'TheHive' case (#772) and performed initial triage. Due to the complexity of the lateral movement detected, I am escalating this to Tier 2 for a deep-dive investigation.

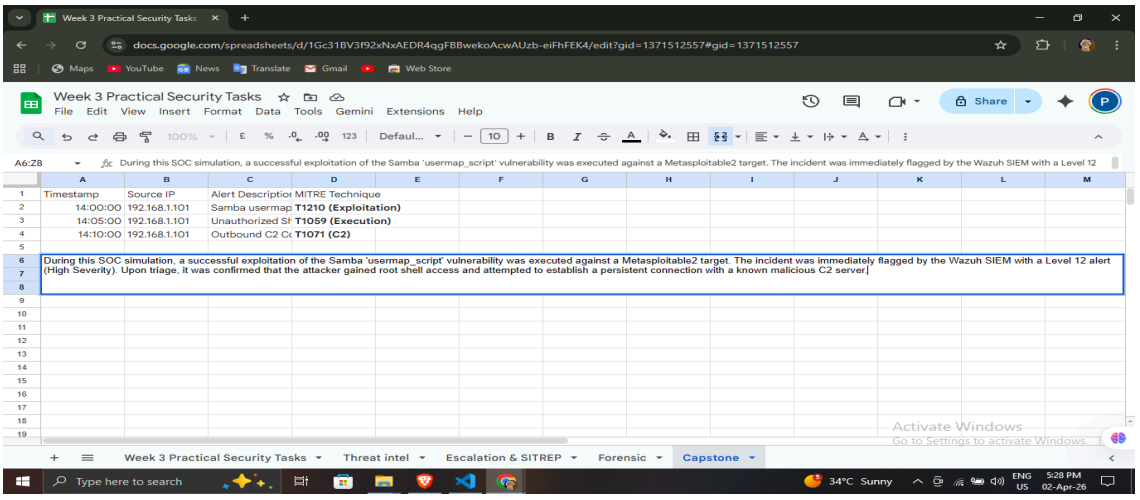
C. Forensics & Capstone Project

Evidence Preservation (Digital Forensics)



Item	Description	Collected By	Date	Hash Value (SHA256)
Network Conne	netstat_output.c	SOC Analyst	2026-04-02	5e884898da28047151d0e56f...
Memory Dump	Server-Y_RAM.r	SOC Analyst	2026-04-02	f7a9e24777ec232127391d...

D. Capstone Project (Full SOC Workflow Simulation)



Timestamp	Source IP	Alert Descriptio	MITRE Technique
14:00:00	192.168.1.101	Samba usermap	T1210 (Exploitation)
14:05:00	192.168.1.101	Unauthorized SI	T1059 (Execution)
14:10:00	192.168.1.101	Outbound C2 Co	T1071 (C2)

During this SOC simulation, a successful exploitation of the Samba 'usermap_script' vulnerability was executed against a Metasploitable2 target. The incident was immediately flagged by the Wazuh SIEM with a Level 12 alert (High Severity). Upon triage, it was confirmed that the attacker gained root shell access and attempted to establish a persistent connection with a known malicious C2 server.